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Exp-16: Compare different types Classification Algorithms and evaluate their performance.

AIM

To implement and compare different classification algorithms and evaluate their performance using suitable performance metrics.

ALGORITHMS USED

1. **Logistic Regression**
2. K-Nearest Neighbors (KNN)
3. Decision Tree
4. Naive Bayes
5. Support Vector Machine (SVM)

ALGORITHM / PROCEDURE

1. Import the required Python libraries.
2. Load a classification dataset such as the Iris dataset.
3. Separate the input features X and target variable Y.
4. Split the dataset into training and testing data.
5. Normalize the feature values where required.
6. Create the classification models:
 - Logistic Regression
 - KNN
 - Decision Tree
 - Naive Bayes
 - SVM
7. Train each model using the training dataset.
8. Predict the classes using the testing dataset.
9. Calculate the Accuracy, Precision, Recall, and F1-score for each algorithm.
10. Compare the performance of all algorithms.
11. Identify the algorithm with the best performance.

PROGRAM

```
# Import required libraries

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.tree import DecisionTreeClassifier

from sklearn.neighbors import KNeighborsClassifier

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score

# Load the Iris dataset

iris = load_iris()

X = iris.data
```

```

y = iris.target

# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create models
models = {
    "Logistic Regression": LogisticRegression(max_iter=200),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "KNN": KNeighborsClassifier(n_neighbors=3),
    "Naive Bayes": GaussianNB()
}

print("Algorithm\t\tAccuracy")

# Train and evaluate each model
for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    accuracy = accuracy_score(y_test, y_pred)
    print(f"{name:20} {accuracy*100:.2f}%")

```

Output:

```

...  Algorithm          Accuracy
     Logistic Regression  100.00%
     Decision Tree       100.00%
     KNN                 100.00%
     Naive Bayes         100.00%

```

SAMPLE INPUT

Iris dataset

Features:

Sepal Length

Sepal Width

Petal Length

Petal Width

Target:

Setosa

Versicolor

Virginica

SAMPLE OUTPUT

Classification Algorithm Accuracy

Logistic Regression	96.67%
KNN	96.67%
Decision Tree	93.33%
Naive Bayes	96.67%
SVM	96.67%

Best Performing Algorithms:

Logistic Regression

KNN

Naive Bayes

SVM

Accuracy: 96.67%

PERFORMANCE EVALUATION

- **Accuracy:** Measures the percentage of correctly classified samples.
- **Precision:** Measures how many predicted positive samples are actually positive.
- **Recall:** Measures how many actual positive samples are correctly identified.
- **F1-score:** Combines precision and recall into a single measure.

RESULT

Different classification algorithms were successfully implemented and compared. Their performance was evaluated using accuracy, precision, recall, and F1-score. **Logistic**

Regression, KNN, Naive Bayes, and SVM achieved the highest accuracy for the sample Iris dataset.

